Quadratics

Chapter Overview

1. Solving quadratic equations
Factorisation, the quadratic formula and substitution.

2. Completing the square
Rewriting and solving quadratic expressions and equations.

3. Quadratics as functions
Inputs, outputs, roots and minimum values.

4. Quadratic graphs
Intercepts, turning points and equations from graphs.

5. The discriminant
How many real roots?

6. Modelling with quadratics
Using and interpreting quadratic models.

Why do we care about quadratics... Solution 3

Quadratic relationships arise in many practical situations.

Summations

$1 + 2 + \cdots + n = \frac{1}{2}n^2 + \frac{1}{2}n$

A sports event awards 8 points, then 7, then 6, and so on. For $1 \leq n \leq 8$ competitors:

$8 + 7 + \cdots + (9 - n) = \frac{n(17 - n)}{2}$

The totals for one, two and three competitors are 8, 15 and 21.

Projectile motion

Under constant acceleration,

$s = ut + \frac{1}{2}at^2$

Here s is displacement, u initial velocity, a acceleration and t time. The expression is quadratic in t.

A projectile moving under gravity alone follows a parabolic path.

Why do we care about quadratics... Solution 4

Hannah's sweets There are n sweets in a bag, of which 6 are orange. Hannah eats two sweets selected at random, without replacement. The probability that both are orange is $1/3$.

Show that $n^2 - n - 90 = 0$.

Hannah's sweets

$$\frac{6}{n} \times \frac{5}{n-1} = \frac{1}{3}$$

$$90 = n(n-1)$$

$$n^2 - n - 90 = 0$$

A product of two expressions involving the same variable often produces a quadratic.

1: Solving Quadratic Equations — Solution 5

Solve $x^2 + 5x = 6$. Two methods are shown below; completing the square is a third method.

1. Factorisation First put the equation in the form $ax^2 + bx + c = 0$.

1. Factorisation

$$x^2 + 5x - 6 = 0$$
$$(x + 6)(x - 1) = 0$$
$$x = -6 \text{ or } 1$$

If a product is zero, at least one factor is zero.

2. Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

2. Quadratic formula

$$a = 1, \quad b = 5, \quad c = -6$$
$$x = \frac{-5 \pm \sqrt{25 + 24}}{2}$$
$$= \frac{-5 \pm 7}{2} = -6 \text{ or } 1$$

1: Solving Quadratic Equations — Solution 6

Solving without expanding

$(x - 1)^2 = 5$

Solving without expanding

$$x - 1 = \pm\sqrt{5} \quad \Rightarrow \quad x = 1 \pm \sqrt{5}$$

Both signs are needed: for example, both 2^2 and $(-2)^2$ equal 4.

Quadratics in disguise

In $x - 6\sqrt{x} + 8$, the expression is quadratic in $\sqrt{x}$. Use substitution or recognise the factors directly.

Substitution Solve $x - 6\sqrt{x} + 8 = 0$, using $y = \sqrt{x} \geq 0$.

Substitution

$$y^2 - 6y + 8 = 0$$
$$(y - 2)(y - 4) = 0$$
$$y = 2 \text{ or } 4 \Rightarrow x = 4 \text{ or } 16$$

Direct factorisation Solve the same equation without substituting.

Direct factorisation

$$(\sqrt{x} - 2)(\sqrt{x} - 4) = 0$$
$$\sqrt{x} = 2 \text{ or } 4$$
$$x = 4 \text{ or } 16$$

Test Your Understanding

Solution 7

1 Solve $(x+3)^2 = x+5$ by factorisation.

1

$$x^2 + 5x + 4 = 0$$
$$(x+4)(x+1) = 0$$
$$x = -4 \text{ or } -1$$

2 Solve $(2x+1)^2 = 5$.

2

$$2x + 1 = \pm\sqrt{5} \quad \Rightarrow \quad x = \frac{-1 \pm \sqrt{5}}{2}$$

3 Solve $\sqrt{x+3} = x - 3$.

3

$$x + 3 = (x-3)^2$$
$$(x-6)(x-1) = 0$$

Check in the original equation: only $x = 6$ works. Squaring can introduce invalid solutions.

4 Solve $2x + \sqrt{x} - 1 = 0$.

4

$$(2\sqrt{x} - 1)(\sqrt{x} + 1) = 0$$

Since $\sqrt{x} \geq 0$, only $\sqrt{x} = 1/2$ is possible. Thus $x = 1/4$.

Calculator check: use polynomial/equation mode with degree 2 for $ax^2 + bx + c = 0$. Show the algebra when the question requires it.

Exercise 2A/2B Pure Y1 Pages 20-22

Solution 8

3(i) Solve $6x^2 - 7 = 11x$.

3(i)

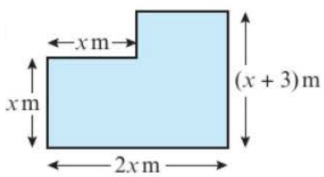
$$(3x - 7)(2x + 1) = 0 \quad \Rightarrow \quad x = \frac{7}{3} \text{ or } -\frac{1}{2}$$

3(h) Solve $5x^2 - 10x^2 = -7 + x + x^2$.

3(h)

$$6x^2 + x - 7 = 0$$
$$(x-1)(6x+7) = 0$$
$$x = 1 \text{ or } -\frac{7}{6}$$

4 This shape has an area of $44\,\text{m}^2$. Find the value of x.



4

$$x^2 + x(x+3) = 44$$
$$(2x + 11)(x - 4) = 0$$

A length is positive, so $x = 4$.

5 Solve $5x + 3 = \sqrt{3x+7}$.

5

$$(5x+3)^2 = 3x + 7$$
$$(25x + 2)(x + 1) = 0$$

The candidates are $-2/25$ and -1. The latter gives $-2 = 2$ in the original equation and is rejected.

Answer: $x = -2/25$.

2(h) Use the quadratic formula to solve $2x^2 + 11x = 5x^2 - 18$. Give answers to three significant figures.

2(h)

$3x^2 - 11x - 18 = 0$

$$x = \frac{11 \pm \sqrt{337}}{6}$$

≈ 4.89 or -1.23

3(e) Solve $(2x + 8)^2 = 100$.

3(e)

$2x + 8 = \pm 10 \quad \Rightarrow \quad x = 1$ or -9

3(f) Solve $6x^2 + 6 = 12x$.

3(f)

$6(x - 1)^2 = 0 \quad \Rightarrow \quad x = 1$

(i) Use $y = \sqrt{x} \geq 0$ to solve $x + 3\sqrt{x} - 1/2 = 0$.

(i)

$2y^2 + 6y - 1 = 0$

$$y = \frac{-3 + \sqrt{11}}{2}$$

$$x = 5 - \frac{3}{2}\sqrt{11}$$

Reject the negative value of y.

(ii)(a) Solve $x + 10\sqrt{x + 2} - 22 = 0$.

(ii)(a)

Let $y = \sqrt{x + 2} \geq 0$, so $x = y^2 - 2$.

$(y + 12)(y - 2) = 0 \quad \Rightarrow \quad y = 2, \; x = 2$

(ii)(b) Solve $x^2 - 4x + \sqrt{2x^2 - 8x - 3} - 9 = 0$.

(ii)(b)

Let $y = \sqrt{2x^2 - 8x - 3} \geq 0$, so $x^2 - 4x = (y^2 + 3)/2$.

$y^2 + 2y - 15 = 0$

$(y + 5)(y - 3) = 0$

$y = 3 \Rightarrow x^2 - 4x = 6$

$x = 2 \pm \sqrt{10}$

2: Completing the Square

Solution 11

Completing the square rewrites a quadratic as $a(x+b)^2+c$. The variable then appears only once, making several tasks easier.

Solving Solve $(x+4)^2-7=0$.

Solving

$x=-4\pm\sqrt{7}$

Sketching

The graph of $y=(x+a)^2+b$ has minimum point $(-a,b)$.

Later integration

In Further Maths, rewriting $x^2-4x+5=(x-2)^2+1$ helps when integrating

$$\int \frac{1}{x^2-4x+5}\,dx$$

Completing the Square Recap

Solution 12

Expand $(x+9)^2=x^2+18x+81$ and $(x-5)^2=x^2-10x+25$.

The number inside the bracket is half the coefficient of x.

1 Complete the square: x^2+12x.

1

$(x+6)^2-36$

Subtract the 36 introduced by expanding the square.

2 Complete the square: x^2+8x.

2

$(x+4)^2-16$

3 Complete the square: x^2-2x.

3

$(x-1)^2-1$

4 Complete the square: x^2-6x+7.

4

$(x-3)^2-9+7=(x-3)^2-2$

$$ax^2+bx+c=a\left(x+\frac{b}{2a}\right)^2+c-\frac{b^2}{4a}\quad(a\neq 0)$$

Think through the steps rather than relying only on memorising this formula.

Completing the Square

Solution 13

Express $2x^2 + 12x + 7$ in completed-square form.

1. Factor out 2

$2(x^2 + 6x) + 7$

2. Complete the square

$2((x + 3)^2 - 9) + 7$

3. Expand

$2(x + 3)^2 - 18 + 7$

4. Simplify

$2(x + 3)^2 - 11$

Express $5 - 3x^2 + 6x$ in completed-square form.

5. Factor out -3

$-3(x^2 - 2x) + 5$

6. Complete the square

$-3((x - 1)^2 - 1) + 5$

7. Expand

$-3(x - 1)^2 + 3 + 5$

8. Simplify

$8 - 3(x - 1)^2$

Test Your Understanding

Solution 14

1 Express $3x^2 - 18x + 4$ as $a(x + b)^2 + c$.

1

$$\begin{aligned} 3(x^2 - 6x) + 4 &= 3((x - 3)^2 - 9) + 4 \\ &= 3(x - 3)^2 - 23 \end{aligned}$$

2 Express $20x - 5x^2 + 3$ as $a - b(x + c)^2$.

2

$$\begin{aligned} -5(x^2 - 4x) + 3 &= -5((x - 2)^2 - 4) + 3 \\ &= 23 - 5(x - 2)^2 \end{aligned}$$

Solving by Completing the Square

Solution 15

Solve $3x^2 - 18x + 4 = 0$ by completing the square.

1. Divide both sides by 3

$$x^2 - 6x + \frac{4}{3} = 0$$

2. Complete the square and solve

$$(x-3)^2 - 9 + \frac{4}{3} = 0$$

$$(x-3)^2 = \frac{23}{3}$$

$$x = 3 \pm \sqrt{\frac{23}{3}}$$

An equation allows division of both sides by the same nonzero number. An expression must retain its value: factor out 3 instead of simply dividing it away.

Exercise 2C/2D Pure Y1 Page 23-24

Solution 16

Complete the square. In each answer, expand the brackets to check that you recover the original expression.

1(a) $x^2 + 4x =$

1(a) $(x+2)^2 - 4$

1(b) $x^2 - 6x =$

1(b) $(x-3)^2 - 9$

1(c) $x^2 - 16x =$

1(c) $(x-8)^2 - 64$

1(d) $x^2 + x =$

1(d) $(x+\frac{1}{2})^2 - \frac{1}{4}$

1(e) $x^2 - 14 =$

1(e) $(x+0)^2 - 14$

2(a) $2x^2 + 16x =$

2(a) $2(x+4)^2 - 32$

2(b) $3x^2 - 24x =$

2(b) $3(x-4)^2 - 48$

2(c) $5x^2 + 20x =$

2(c) $5(x+2)^2 - 20$

2(d) $2x^2 - 5x =$

2(d) $2(x-\frac{5}{4})^2 - \frac{25}{8}$

2(e) $8x - 2x^2 =$

2(e) $8 - 2(x-2)^2$

Exercise 2C/2D Pure Y1 Page 23-24 — Solution 17

4 Given $x^2 + 3x + 6 = (x + a)^2 + b$, find a, b.

4

$x^2 + 3x + 6 = (x + \frac{3}{2})^2 + \frac{15}{4}$

Thus $a = 3/2$, $b = 15/4$.

5 Write $2 + 0.8x - 0.04x^2$ as $A - B(x + C)^2$.

5

$6 - 0.04(x - 10)^2$

$A = 6,\ B = 0.04,\ C = -10.$

Proof Complete the square to show that $x^2 + 2bx + c = 0$ has solutions $x = -b \pm \sqrt{b^2 - c}$.

Proof

$$(x + b)^2 - b^2 + c = 0$$
$$(x + b)^2 = b^2 - c$$
$$x = -b \pm \sqrt{b^2 - c}$$

Real solutions require $b^2 - c \geq 0$.

Proving the Quadratic Formula — Solution 18

Given $ax^2 + bx + c = 0$, with $a \neq 0$, prove the quadratic formula by completing the square.

Proof

$$x^2 + \frac{b}{a}x + \frac{c}{a} = 0$$
$$\left(x + \frac{b}{2a}\right)^2 = \frac{b^2 - 4ac}{4a^2}$$
$$(2ax + b)^2 = b^2 - 4ac$$
$$2ax + b = \pm\sqrt{b^2 - 4ac}$$
$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

3: Functions

Solution 19

A function gives a rule that assigns one output to each permitted input. It is a formal version of a number machine.

input x $\longrightarrow$ f $\longrightarrow$ output $f(x)$

Example The function doubles its input. Write its rule.

Example

$f(x) = 2x$

f is the name of the function; $f(x)$ is its output for input x.

3: Functions

Solution 20

For $f(x) = x^2$, example inputs and outputs are:

x	-1	0	1.7	2	3.1
$f(x)$	1	0	2.89	4	9.61

Domain and range What do these terms mean?

Domain and range

The domain is the set of permitted inputs. The range is the set of outputs achieved.

With domain $x \in \mathbb{R}$, the function $f(x) = x^2$ has range $f(x) \geq 0$.

Roots What is a root (or zero) of a function?

Roots

A value of x for which $f(x) = 0$.

For $f(x) = x^2$, the only root is $x = 0$.

$x \in \mathbb{R}$ reads: x is a member of the set of real numbers.

Examples

Solution 21

Let $f(x) = x^2 - 3x$ and $g(x) = x + 5$, with $x \in \mathbb{R}$.

(a) Find $f(-4)$.

(a)

$f(-4) = (-4)^2 - 3(-4) = 28$

(b) Find x for which $f(x) = g(x)$.

(b)

$x^2 - 4x - 5 = (x - 5)(x + 1) = 0$

$x = 5$ or $x = -1$.

(c) Find the roots of f.

(c)

$x(x - 3) = 0 \quad \Rightarrow \quad x = 0$ or 3

(d) Find the root of g.

(d)

$x + 5 = 0 \quad \Rightarrow \quad x = -5$

Examples

Solution 22

Find the minimum value of $f(x) = x^2 - 6x + 2$, and the value of x where it occurs.

Trying inputs suggests an answer:

x	1	2	3	4
$f(x)$	-3	-6	-7	-6

But a few examples do not prove a minimum.

Complete the square

$f(x) = (x - 3)^2 - 7$

Since $(x - 3)^2 \geq 0$, the minimum is -7, occurring at $x = 3$.

General rule

For $f(x) = (x + a)^2 + b$, the minimum is b, at $x = -a$.

Quickfire Questions

Solution 23

For each function, complete the square, state its minimum or maximum, and give the input where it occurs.

1

$f(x) = x^2 + 4x + 9$

1

$f(x) = (x+2)^2 + 5$

Minimum 5 at $x = -2$.

2

$f(x) = x^2 - 10x + 21$

2

$f(x) = (x-5)^2 - 4$

Minimum -4 at $x = 5$.

3

$f(x) = 10 - x^2$

3

Already in completed-square form.

Maximum 10 at $x = 0$.

4

$f(x) = 8 - x^2 + 6x$

4

$f(x) = 17 - (x-3)^2$

Maximum 17 at $x = 3$.

Test Your Understanding

Solution 24

1 Find the minimum of $2x^2 + 12x - 5$, and where it occurs.

1

$2x^2 + 12x - 5 = 2(x+3)^2 - 23$

Minimum -23 at $x = -3$.

2 Find the roots of $2x^2 + 3x + 1$.

2

$(2x+1)(x+1) = 0 \quad \Rightarrow \quad x = -\frac{1}{2}, -1$

3 Find the real roots of $x^4 - x^2 - 6$.

3

$(x^2 - 3)(x^2 + 2) = 0$

Only $x^2 = 3$ gives real solutions: $x = \pm\sqrt{3}$.

Exercise 2E Pure Y1 Pages 26-27 Solution 25

2 Given $f(x) = x^2 - 2x$, find a when $f(a) = 8$.

2

$a^2 - 2a - 8 = (a-4)(a+2) = 0$

$a = 4$ or $a = -2$.

6 Complete the square for $f(x) = x^2 - 2x + 2$. Explain why $f(x) > 0$ for all real x.

6

$f(x) = (x-1)^2 + 1 \geq 1 > 0$

8 Find the roots of $f(x) = 3^{2x} - 28 \cdot 3^x + 27$.

8

$(3^x - 27)(3^x - 1) = 0$

$3^x = 27$ or $1 \Rightarrow x = 3$ or 0

4: Quadratic Graphs Solution 26

Sketching a quadratic

Find and label the key features. A sketch does not require a uniform numerical scale.

1. Roots

Solve $ax^2 + bx + c = 0$ to find any x-intercepts.

2. Vertical intercept

Put $x = 0$: the point is $(0, c)$.

3. Shape

If $a > 0$, the parabola opens upwards. If $a < 0$, it opens downwards.

4. Turning point

Complete the square: $y = a(x-h)^2 + k$ has turning point (h, k).

The axis of symmetry is $x = h$. Draw a smooth curve through the labelled points.

Example

Solution 27

Sketch $y = x^2 + 3x - 4$, showing the intercepts and turning point.

1. Find key points

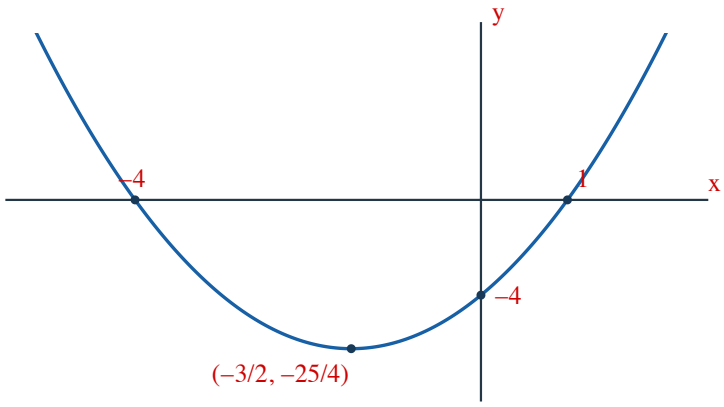
$(x + 4)(x - 1) = 0$

Roots: $-4, 1$. Vertical intercept: $(0, -4)$.

$y = (x + \frac{3}{2})^2 - \frac{25}{4}$

Minimum: $(-3/2, -25/4)$.

2. Sketch



Example

Solution 28

Sketch $y = 4x - 2x^2 - 3$.

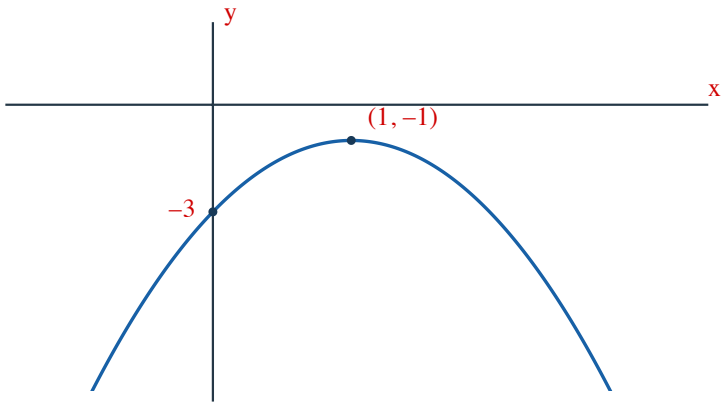
1. Key features

$y = -2(x - 1)^2 - 1$

Maximum: $(1, -1)$; vertical intercept: $(0, -3)$.

There are no real roots. The axis of symmetry is $x = 1$.

2. Sketch



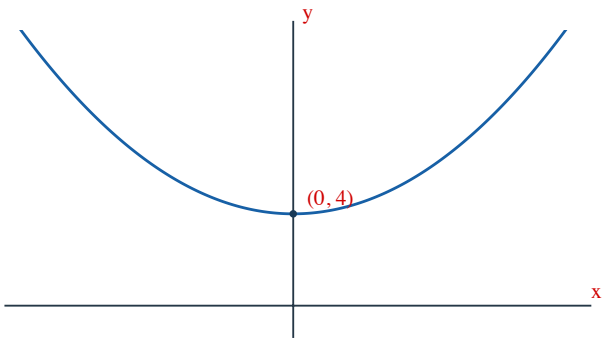
Test Your Understanding

Solution 29

Sketch each quadratic, labelling the intercepts and turning point.

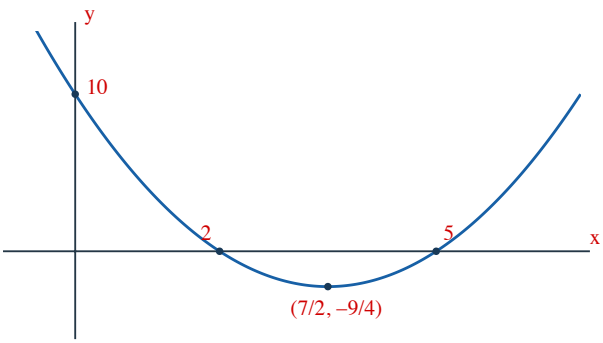
$y = x^2 + 4$

a



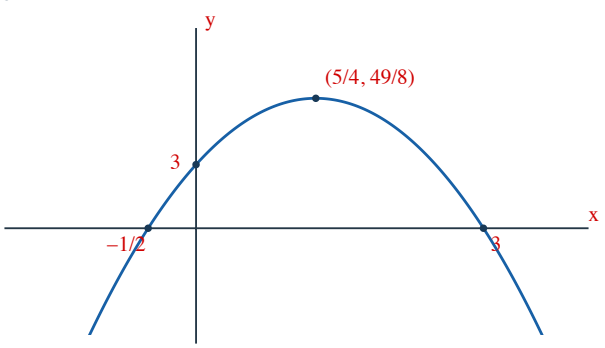
$y = x^2 - 7x + 10$

b



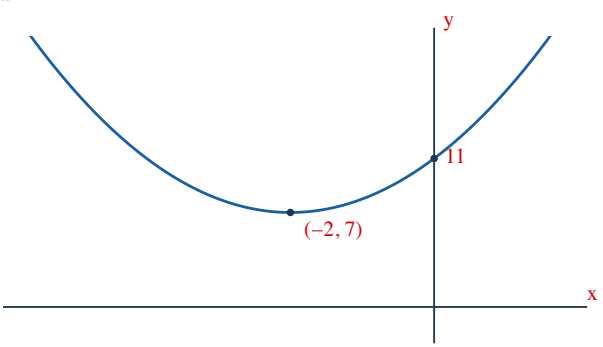
$y = 5x + 3 - 2x^2$

c



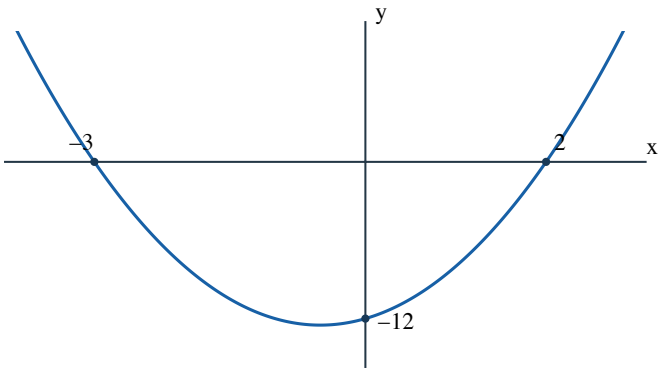
$y = x^2 + 4x + 11$

d



Determining the Equation using a Graph

Solution 30



Question Find the equation in the form $y = ax^2 + bx + c$.

Question

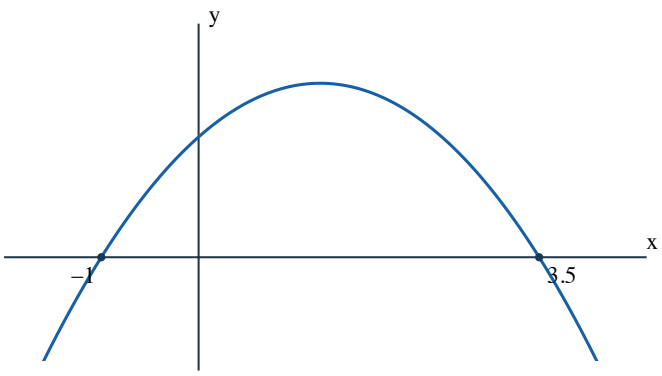
$$y = a(x + 3)(x - 2)$$

$$-12 = -6a \quad \Rightarrow \quad a = 2$$

$$y = 2x^2 + 2x - 12$$

Further Example

Solution 31



Question Find one possible equation with integer coefficients.

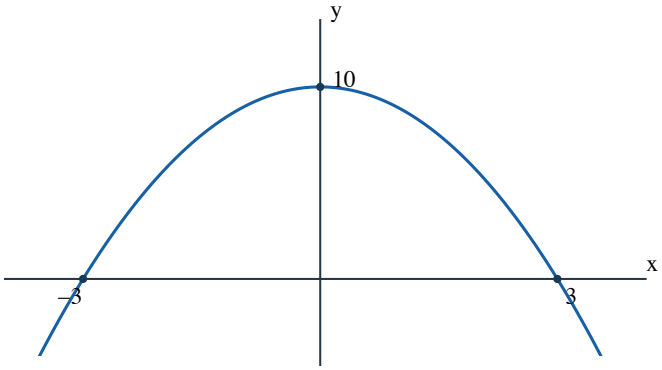
Question

$y = -(x+1)(2x-7) = -2x^2 + 5x + 7$

The intercept on the vertical axis is not specified. Any positive multiple has the same roots and downward opening, so the equation is not unique.

Test Your Understanding

Solution 32



Question Determine an equation of this graph.

Question

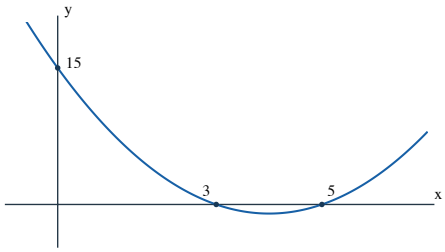
$y = a(x+3)(x-3)$

$10 = -9a \quad \Rightarrow \quad a = -\frac{10}{9}$

$y = \frac{10}{9}(x+3)(3-x)$

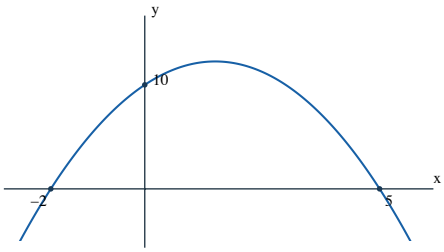
Exercise 2F Pure Y1 Page 30 — Solution 33

Find the equation $y = ax^2 + bx + c$ of each graph.



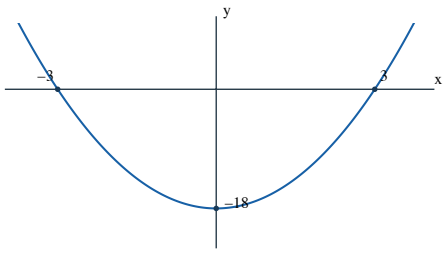
a

$y = (x-3)(x-5) = x^2 - 8x + 15$



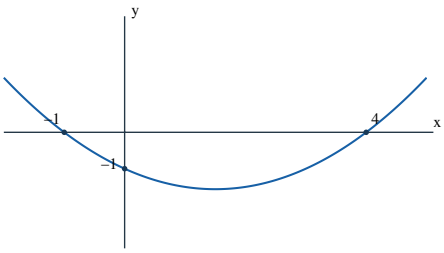
b

$y = -(x+2)(x-5) = -x^2 + 3x + 10$



c

$y = 2(x+3)(x-3) = 2x^2 - 18$



d

$y = \frac{1}{4}(x+1)(x-4) = \frac{1}{4}x^2 - \frac{3}{4}x - 1$

Exercise 2F Pure Y1 Page 30 — Solution 34

3 The graph $y = ax^2 + bx + c$ has minimum $(5, -3)$ and passes through $(4, 0)$. Find a, b, c.

3

$y = a(x-5)^2 - 3$

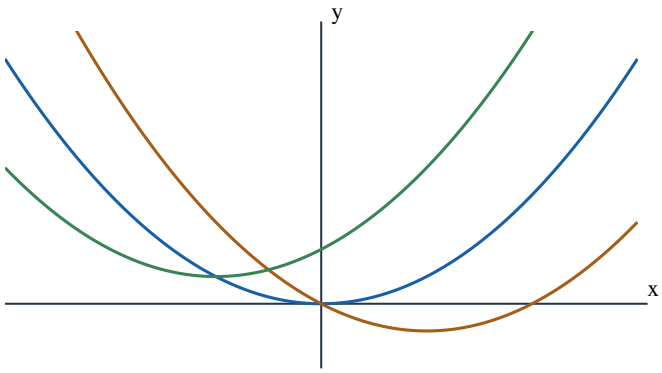
$0 = a(4-5)^2 - 3 \Rightarrow a = 3$

$y = 3x^2 - 30x + 72$

$a = 3,\ b = -30,\ c = 72.$

Extension · MAT 2003 1H

Into how many regions do $y = x^2$, $y = x^2 - 2x$, $y = x^2 + 2x + 2$ divide the plane?



Extension

The first parabola makes 2 regions. The second crosses it once and adds 2. The third crosses each previous curve at distinct points and adds 3.

Total: $2 + 2 + 3 = 7$ regions.

Starter

Solution 35

How many distinct real solutions does each equation have?

1

$x^2 - 12x + 36 = 0$

1

$(x-6)^2 = 0$

One distinct solution: $x = 6$.

2

$x^2 + x + 3 = 0$

2

$b^2 - 4ac = 1 - 12 = -11$

No real solutions.

3

$x^2 - 2x - 1 = 0$

3

$x = 1 \pm \sqrt{2}$

Two distinct real solutions.

5: The Discriminant

Solution 36

$ax^2 + bx + c = 0, \quad a \neq 0 \qquad x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

The discriminant is $D = b^2 - 4ac$.

No real roots What condition is needed?

No real roots

$D < 0$

Equal roots What condition is needed?

Equal roots

$D = 0$

Adding or subtracting zero gives the same root.

Two distinct real roots What condition is needed?

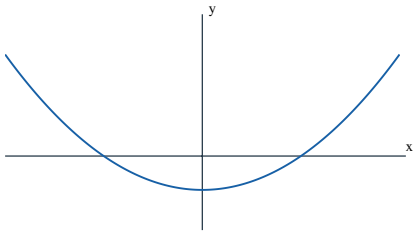
Two distinct real roots

$D > 0$

Roots are the values of x that make the quadratic equal to zero.

Two distinct real roots

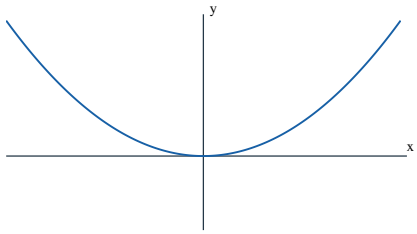
$D > 0$



The graph crosses the horizontal axis twice.

Equal roots

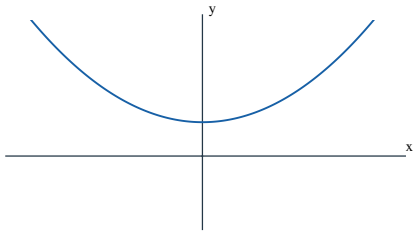
$D = 0$



The graph touches the horizontal axis at its turning point.

No real roots

$D < 0$



The graph does not meet the horizontal axis.

Why say "equal roots"?

$x^2 - 12x + 36 = (x - 6)^2$

The root 6 occurs twice as a factor. There is one distinct root, with multiplicity 2.

Fundamental theorem of algebra

Every polynomial of degree $n \geq 1$ has exactly n complex roots when counted with multiplicity.

The degree is the highest power with a nonzero coefficient. A quadratic has degree 2.

Thus a quadratic always has two complex roots counting multiplicity, even when it has no real roots.

An example of factorisation over the reals

$$x^4 + a^4 = (x^2 + \sqrt{2}ax + a^2) \times (x^2 - \sqrt{2}ax + a^2)$$

Expanding gives $(x^2 + a^2)^2 - 2a^2x^2 = x^4 + a^4$.

Every real polynomial can be factored into real linear and quadratic factors.

Quickfire Questions

Solution 39

Find the discriminant and state the number of distinct real roots.

1

$x^2 + 3x + 4 = 0$

1

$D = -7$

0 distinct real roots.

2

$x^2 - 4x + 1 = 0$

2

$D = 12$

2 distinct real roots.

3

$x^2 - 4x + 4 = 0$

3

$D = 0$

1 distinct real root.

4

$2x^2 - 6x - 3 = 0$

4

$D = 60$

2 distinct real roots.

5

$x - 4 - 3x^2 = 0$

5

$D = -47$

0 distinct real roots.

6

$1 - x^2 = 0$

6

$D = 4$

2 distinct real roots.

Problems involving the discriminant

Solution 40

The equation $x^2 + 2px + (3p + 4) = 0$ has equal roots, where $p > 0$. Find p, then solve the equation.

1. Set the discriminant to zero

$(2p)^2 - 4(3p + 4) = 0$

2. Find p

$p^2 - 3p - 4 = (p - 4)(p + 1) = 0$

Since $p > 0$, $p = 4$.

3. Find the repeated root

$x^2 + 8x + 16 = (x + 4)^2 = 0 \quad \Rightarrow \quad x = -4$

Exercise 2G Pure Y1 Page 32 — Solution 41

1 Find the values of k for which $x^2 + 6x + k = 0$ has two distinct real roots.

1

$36 - 4k > 0 \quad \Rightarrow \quad k < 9$

2 Find t if $2x^2 - 3x + t = 0$ has a repeated root.

2

$9 - 8t = 0 \quad \Rightarrow \quad t = \frac{9}{8}$

Exercise 2G Pure Y1 Page 32 — Solution 42

1 The equation $sx^2 + 8x + s = 0$ has equal roots, with $s > 0$. Find s.

1

$64 - 4s^2 = 0 \quad \Rightarrow \quad s = 4$

2 Find the values of k for which $3x^2 - 4x + k = 0$ has no real roots.

2

$16 - 12k < 0 \quad \Rightarrow \quad k > \frac{4}{3}$

Exercise 2G Pure Y1 Page 32 Solution 43

6 $g(x) = x^2 + 3px + (14p - 3)$ has equal roots, and p is an integer. Find p and the root.

6

$9p^2 - 56p + 12 = 0$

$(9p - 2)(p - 6) = 0$

The integer solution is $p = 6$. Then $g(x) = (x + 9)^2$, with root -9.

7 Show that $2x^2 + (k + 4)x + k = 0$ has two distinct real roots for every real k.

7

$D = (k + 4)^2 - 8k = k^2 + 16 > 0$

Test Your Understanding Solution 44

1 Find k when $x^2 + 5kx + (10k + 5) = 0$ has equal roots.

1

$25k^2 - 40k - 20 = 0$

$5(5k + 2)(k - 2) = 0$

$k = -\frac{2}{5}$ or 2

2 Find k when $x^2 + 6x + k = 0$ has two distinct real roots.

2

$36 - 4k > 0 \quad \Rightarrow \quad k < 9$

Exercise 2G Pure Y1 Page 32 — Solution 45

1. MAT 2011 1B A rectangle has perimeter P and area A. Which must hold?

$P^3 > A$; $A^2 > 2P + 1$; $P^2 \geq 16A$; $PA > A + P$

1. MAT 2011 1B

With side lengths $x, y > 0$, $A = xy$ and $P = 2x + 2y$.

$2x^2 - Px + 2A = 0$

Real side lengths require $P^2 - 16A \geq 0$. Thus $P^2 \geq 16A$.

2. MAT 2006 1B How many real roots does $(2 + x - x^2)^2 = 16$ have?

2. MAT 2006 1B

Consider $2 + x - x^2 = 4$ and $2 + x - x^2 = -4$.

The first gives $x^2 - x + 2 = 0$, with negative discriminant. The second gives $(x + 2)(x - 3) = 0$.

Two distinct real roots: $-2, 3$.

3. MAT 2009 1C For which real c does $x^4 = (x - c)^2$ have four real roots, counting repetitions?

3. MAT 2009 1C

The factors give $x^2 - x + c = 0$ and $x^2 + x - c = 0$. Both need nonnegative discriminants:

$$1 - 4c \geq 0, \quad 1 + 4c \geq 0 \quad \Rightarrow \quad -\frac{1}{4} \leq c \leq \frac{1}{4}$$

Modelling — Solution 46

A spear is thrown from a tower over level ground. Its height in metres after t seconds is modelled by

$h(t) = 12.25 + 14.7t - 4.9t^2, \quad t \geq 0$

A mathematical model describes a situation using simplifying assumptions.

(a) Interpret 12.25.

(a)

The initial height is $h(0) = 12.25$ m: the height of the tower.

(b) When does the spear hit the ground?

(b)

Solve $h(t) = 0$: $t \approx -0.679$ or 3.68.

Reject the negative time; the spear lands after 3.68 s (3 s.f.).

(c) Write $h(t)$ as $A - B(t - C)^2$.

(c)

$h(t) = 23.275 - 4.9(t - 1.5)^2$

(d) Find the maximum height and when it is reached.

(d)

The maximum height is 23.275 m, reached 1.5 s after release.

Exercise 2H Pure Y1 Page 34 — Solution 47

A car model predicts fuel economy y mpg at speed x mph:

$y = -0.01x^2 + 0.975x + 16, \quad x > 0$

(a) Find the two speeds giving 32.5 mpg.

(a)

$0.01x^2 - 0.975x + 16.5 = 0$

$x \approx 21.8$ mph or 75.7 mph (3 s.f.).

(b) Rewrite as $A - B(x - C)^2$.

(b)

$y = 39.765625 - 0.01(x - 48.75)^2$

(c) At what speed is fuel economy greatest?

(c)

At $x = 48.75$ mph, the squared term is zero and y is greatest.

(d) Calculate the prediction at 120 mph and comment on validity.

(d)

$y = -0.01(120)^2 + 0.975(120) + 16 = -11$

Negative mpg is impossible. The model is not valid at very high speeds.

Exercise 2H Pure Y1 Page 34 — Solution 48

Grain yield g, in tonnes per hectare, is modelled using fertiliser f, in kilograms per hectare:

$g = 6 + 0.03f - 0.00006f^2$

(a) Find the yield without fertiliser.

(a)

At $f = 0$, the yield is 6 tonnes per hectare.

(b) A farmer currently uses 20 kg per hectare. Find the smallest additional amount needed to increase yield by 1 tonne per hectare.

(b)

$$g(20) = 6.576$$

$$6 + 0.03f - 0.00006f^2 = 7.576$$

$$f \approx 59.6 \text{ or } 440.4$$

The smaller increase is $59.6 - 20 = 39.6$ kg per hectare (1 d.p.).

Exercise 2H Pure Y1 Page 34 — Solution 49

A stadium has 25,000 seats. At £30 per ticket, 10,000 tickets sell. Each £1 reduction sells 1,000 more tickets. Model sales by $t = M - 1000p$, where £p is the ticket price.

(a) Find M.

(a)

$10000 = M - 1000(30) \quad \Rightarrow \quad M = 40000$

(b) Revenue is $r = p(M - 1000p)$. Write it as $A - B(p - C)^2$.

(b)

$r = 400000 - 1000(p - 20)^2$

(c) Find the price that maximises revenue.

(c)

Charge £20, giving revenue £400,000.

Sales are 20,000 tickets, within the 25,000-seat capacity.

Would you like $1,000,000 for finding roots? — Solution 50

Roots are inputs that give an output of zero. Finding roots can also lead to major research problems.

For real $s > 1$, the Riemann zeta function can be defined by

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}$$

$$\zeta(2) = 1 + \frac{1}{2^2} + \frac{1}{3^2} + \cdots$$

The function extends to complex inputs through a more advanced definition.

The Riemann hypothesis

It predicts that every nontrivial zero of the extended zeta function has real part $1/2$.

This is a Millennium Prize Problem with a US$1 million prize for a qualifying solution. It connects zeros of a function with the distribution of prime numbers.

Read the Clay Mathematics Institute explanation